LEAN CANVAS — FoodLink Agent (Autonomous AI Food Rescue Agent)

AI/ML Upgrades: Surplus Forecasting · Arrhenius Shelf-Life · Smart Matching · VRP · Anomaly Detection
Stack: AWS Bedrock AgentCore · DynamoDB ACID · Scikit-Learn · Regional i18n · **Format:** Single-Page A4

1. PROBLEM

- **Reactive Blindspot:** Shelters get notified only when food is already surplus, lacking 24h advance preparation notice.
- **Perishable Spoilage Window:** Cooked meals spoil in 3–4h at ambient danger-zone temps (>30°C) without thermal routing.
- **Sub-Optimal Single Trips:** 1-pickup/1-dropoff missions create high volunteer fuel costs & transit delays.
- **Coordinator Burnout:** Non-profits waste 3–5h daily juggling chaotic WhatsApp calls & Excel sheets.
- **Fraud & Fake Donations:** Bot reports, spam quantities, or repeated cancellations waste driver dispatch.

EXISTING ALTERNATIVES

- Chaotic WhatsApp / Telegram broadcast groups
  - Manual calling chains & unverified drop-ins
  - Dumping wholesome edible food into city landfills

4. SOLUTION (AI/ML POWERED)

- **Surplus Forecasting Model:** 24h advance notice using Ridge regression on historical day-of-week & event cycles.
- **Physics-Informed Shelf-Life:** Arrhenius thermal decay model + strict <60m coordinator escalation override.
- **Smart Hybrid Matching:** Blends deterministic proximity/dietary rules with ML-predicted shelter acceptance prob.
- **Multi-Stop VRP Optimizer:** Heuristic solver batching 2–3 pickups & dropoffs minimizing spoilage penalties.
- **Anomaly Detection:** Isolation Forest flags bulk spikes (>3σ), unrealistic claims & bot velocity.
- **Indore Hotspot Map & i18n:** Ward-level surplus clustering (Sarafa, 56 Dukan) with bilingual English/Hindi.

8. KEY METRICS

- **Forecast Accuracy (MAPE):** >88% surplus volume accuracy 24h in advance.
- **Total Food Rescued:** Total kg diverted & USDA meals equivalent (×2.0).
- **Autonomous Match Latency:** <0.8s dispatch with zero IDOR leaks.
- **VRP Multi-Stop Batch Factor:** 1.8 stops/trip (+35% fleet fuel efficiency).
- **Spoilage Reduction:** >98.2% edible food rescued within safe window.
- **Anomaly Precision:** 99.4% fraudulent reports caught before dispatch.

3. UNIQUE VALUE PROPOSITION

Autonomous, Predictive Food Rescue Coordination combining 24h advance surplus forecasting, physics-based shelf-life safety, multi-stop VRP batch routing, and cryptographic zero-leak privacy.

- **For Donors:** 30-sec zero-login intake with AI shelf-life auto-calculator, fraud protection & verified CSR ESG receipts.
- **For Shelters:** 24-hour advance intake forecasting + ML-personalized dietary matching—no surprise food dumps.
- **For Volunteers:** Multi-stop VRP batch routes minimizing mileage & food decay while maximizing rescue impact.
- **For Coordinators:** Transforms frantic dispatchers into strategic exception managers via automated triage.

HIGH-LEVEL CONCEPT

- *"Autonomous Uber + DoorDash logistics engine, with Arrhenius food safety and ML predictive intelligence for zero-waste."*

5. UNFAIR ADVANTAGE

- **Arrhenius Spoilage + VRP Formulation:** Proprietary joint optimization of bacterial kinetics and multi-stop transit.
- **Hybrid Deterministic + ML Ranker:** Combines hard food safety & capacity rules with learned NGO acceptance behavior.
- **DynamoDB ACID Conditional Locks:** Mathematical guarantee against double allocation of shelter capacity.
- **Zero-IDOR Cryptographic Privacy:** Salted SHA-256 tokens mask donor/shelter PII & exact GPS coordinates.
- **Dual Anomaly Guardrails:** Isolation Forest + Z-score blocks fake spam before volunteer dispatch.

9. CHANNELS

- **Predictive Advance SMS/WhatsApp:** Proactive 24h intake notifications to NGO directors.
- **Kitchen QR Action Stickers:** 1-scan laminated badges on banquet & restaurant prep tables.
- **Municipal Food Alliances:** Indore IMC Ward integration & urban food security initiatives.
- **Civic EV & Bike Fleets:** University cycling clubs & volunteer two-wheeler courier groups.
- **Public Impact Dashboard:** Real-time public ticker showing diverted kg & CO2 saved.

2. CUSTOMER SEGMENTS

- **Commercial Food Donors:** Restaurants, wedding banquet halls, student mess kitchens, cloud kitchens, bakeries.
- **Recipient Partners:** Verified homeless shelters, community kitchens, rain baseras, orphanages, elder homes.
- **Transit Volunteers:** EV two-wheeler couriers, community drivers, and local cargo cyclists.
- **Relief Coordinators:** Municipal & NGO directors managing regional zero-hunger programs.

EARLY ADOPTERS

- High-waste wedding corridors (e.g. Vijay Nagar Banquets).
  - Night food markets with closing surplus (Sarafa Bazaar).
  - Dense student hostel mess clusters (Bhawarkua, Indore).

7. COST STRUCTURE

- **Cloud & ML Compute:** AWS Bedrock AgentCore, Lambda serverless, Scikit-Learn training pipelines, DynamoDB ACID tables.
  - **Geospatial & VRP Engine:** Location matrix routing, distance API calls, and turn-by-turn waypoint computation.
- **Communication Infrastructure:** Amazon SNS, Twilio WhatsApp / SMS proactive advance alert gateways.
  - **Vetting & Field Operations:** NGO health inspection verification, driver thermal insulated bags, and coordinator training.

6. REVENUE STREAMS (FINANCIAL SUSTAINABILITY)

- **Predictive Kitchen Waste Analytics (SaaS):** Paid B2B analytics portal helping restaurants cut 15–20% over-prep food waste.
  - **Enterprise ESG / CSR Subscriptions:** Automated Scope-3 greenhouse gas & tax diversion compliance certificates.
- **Municipal Smart-City Waste Grants:** City funding (e.g. Indore IMC) for landfill methane mitigation & waste diversion.
  - **Philanthropic Endowments:** Zero-hunger civic tech grants and sustainable food system foundations.